WORKED SOLUTION

STEP III 2004, Question 11

Mechanics • Two-Body Repulsion

STEP III 2004, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.6, *Two Bodies Become One*: https://beyondstep.keithmcnulty.org/ch10/two-body.html

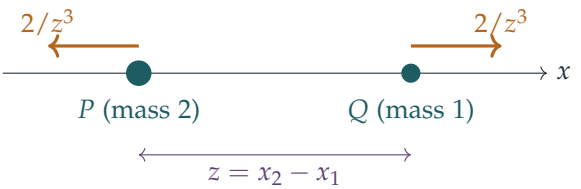
The key idea

Two particles pushing each other apart give two equations of motion, and two clever combinations of them do all the work. The *difference* $x_2 - x_1$ gives a single equation for the separation z, which the $v\,\mathrm{d}v/\mathrm{d}z$ trick integrates once (it is really energy conservation). The *weighted sum* $2x_1 + x_2$ is three times the centre of mass, and since the forces are internal it feels no net force at all.

Solution

The equation for z

Model P and Q as particles on a line, with x increasing from P towards Q (since $x_2 > x_1$). The repulsion pushes P in the negative direction and Q in the positive direction, each with magnitude $2/z^3$.



Newton's second law for each particle gives

$$2\ddot{x}_1 = -\frac{2}{z^3}, \qquad \ddot{x}_2 = \frac{2}{z^3}.$$

So $\ddot{x}_1 = -1/z^3$ and, subtracting,

$$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \ddot{x}_2 - \ddot{x}_1 = \frac{2}{z^3} + \frac{1}{z^3} = \frac{3}{z^3},$$

as required.

Solving for z

First the initial conditions. At $t = 0$ we have $z = 1$. P moves towards Q, that is, in the positive direction, so $\dot{x}_1 = 1$, while $\dot{x}_2 = 0$. Hence $v = \dot{z} = 0 - 1 = -1$ at $t = 0$: the gap is closing.

With $\ddot{z} = v\,\mathrm{d}v/\mathrm{d}z$, the equation becomes

$$v\frac{\mathrm{d}v}{\mathrm{d}z} = \frac{3}{z^3} \quad \Longrightarrow \quad \tfrac{1}{2}v^2 = -\frac{3}{2z^2} + C.$$

At $z = 1$, $v^2 = 1$, so $\frac{1}{2} = -\frac{3}{2} + C$ and $C = 2$. Therefore

$$v^2 = 4 - \frac{3}{z^2}, \qquad \text{or} \qquad z^2\Big(\frac{\mathrm{d}z}{\mathrm{d}t}\Big)^2 = 4z^2 - 3.$$

The left-hand side is $\big(\frac{1}{2}\,\mathrm{d}(z^2)/\mathrm{d}t\big)^2$, which suggests putting $u = \sqrt{4z^2 - 3} \geq 0$. Then $u^2 = 4z^2 - 3$, so $u\,\dot{u} = 4z\dot{z}$, and the equation above says $(z\dot{z})^2 = u^2$, that is, $z\dot{z} = \pm u$. Hence

$$\dot{u} = \frac{4z\dot{z}}{u} = \pm 4.$$

Initially $u = 1$ and $z\dot{z} = -1 < 0$, so we take the minus sign: $u = 1 - 4t$.

Ask yourself What happens when u reaches 0, and does the sign of $\dot{z}$ stay negative?

No. At $t = \frac{1}{4}$ we get $u = 0$, so $z = \sqrt{3}/2$ and $v = 0$: the particles are momentarily at their closest. There $\ddot{z} = 3/z^3 > 0$, so z starts to increase, $\dot{z}$ becomes positive and the plus sign takes over, giving $u = 4t - 1$ for $t \geq \frac{1}{4}$ (continuous at $t = \frac{1}{4}$). In both cases $u^2 = (4t - 1)^2$, so for all $t \geq 0$

$$4z^2 - 3 = 16t^2 - 8t + 1 \quad \Longrightarrow \quad z^2 = 4t^2 - 2t + 1,$$

and since $z > 0$,

$$z = \sqrt{4t^2 - 2t + 1}.$$

This is the required result. (Note $4t^2 - 2t + 1 = 4(t - \frac{1}{4})^2 + \frac{3}{4} > 0$, so the square root is always defined.)

Finding x_1 and x_2

From the two equations of motion,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}(2x_1 + x_2) = 2\ddot{x}_1 + \ddot{x}_2 = -\frac{2}{z^3} + \frac{2}{z^3} = 0.$$

This is conservation of momentum: the forces are equal and opposite. So $2x_1 + x_2$ is linear in t. At $t = 0$ it equals $2 \cdot 0 + 1 = 1$, and its rate of change is $2\dot{x}_1 + \dot{x}_2 = 2 \cdot 1 + 0 = 2$. Hence

$$2x_1 + x_2 = 2t + 1, \qquad x_2 - x_1 = \sqrt{4t^2 - 2t + 1}.$$

Subtracting the second from the first gives $3x_1$, and adding twice the second to the first gives $3x_2$:

$$x_1 = \frac{2t + 1 - \sqrt{4t^2 - 2t + 1}}{3}, \qquad x_2 = \frac{2t + 1 + 2\sqrt{4t^2 - 2t + 1}}{3}.$$

Checks

- At $t = 0$ these give $x_1 = 0$ and $x_2 = 1$. Differentiating, $\dot{x}_1(0) = \frac{1}{3}\big(2 - \frac{1}{2} \cdot (-2)\big) = 1$ and $\dot{x}_2(0) = \frac{1}{3}(2 - 2) = 0$, as required.
- A quicker route to z: with $w = z^2$, $\ddot{w} = 2\dot{z}^2 + 2z\ddot{z} = 2\big(4 - 3/z^2\big) + 6/z^2 = 8$, so $w = 4t^2 - 2t + 1$ from $w(0) = 1$ and $\dot{w}(0) = 2z\dot{z} = -2$. This sidesteps the sign worry altogether.
- As $t \to \infty$, $z \approx 2t - \frac{1}{2}$, so $x_1 \to \frac{1}{2}$ and $\dot{x}_2 \to 2$: P ends up at rest and Q carries all the momentum ($1 \times 2 = 2 \times 1$) and all the energy. The potential energy is $1/z^2$, so the energy at the start is $1 + 1 = 2$ and at the end it is $\frac{1}{2} \cdot 1 \cdot 2^2 = 2$.
- Stepping the original equations numerically (Runge–Kutta, step 10^{-4}) up to $t = 5$ matches the formulae for x_1, x_2 and z to within 10^{-11}, with closest approach $z = 0.8660\ldots = \sqrt{3}/2$.

Common slips

- Taking $\dot{z}(0) = +1$. P is moving towards Q, so the gap is shrinking.
- Writing $\mathrm{d}z/\mathrm{d}t = -\sqrt{4 - 3/z^2}$ for all time and missing the turning point at $t = \frac{1}{4}$.
- Forgetting the mass 2 in P's equation, which gives $\ddot{z} = 4/z^3$.

Beyond STEP

This is the two-body problem in miniature: the centre of mass moves uniformly, and the relative coordinate behaves like a single particle of *reduced mass* $m_1 m_2/(m_1 + m_2) = \frac{2}{3}$ feeling the same force, since $\frac{2}{3}\ddot{z} = 2/z^3$. The same split reduces planetary motion to a single orbit. The inverse-cube force is special too: z^2 turns out to be exactly quadratic in t, a hidden symmetry of the $1/z^2$ potential studied in conformal mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats reduced mass and two-body motion carefully.

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